

A Halal Path Forward: How Decentralized Exchanges Enable Shariah-Compliant Perpetual Futures Trading for the First Time

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Abstract

This paper documents a critical discovery: for the first time in modern finance, perpetual futures contracts can be structured in a manner compatible with Islamic Shariah principles. Prior to this finding, perpetual futures were universally deemed *haram* (prohibited) by Islamic scholars due to explicit interest-bearing components embedded in their funding mechanisms. However, the emergence of decentralized exchange (DEX) technology has inadvertently created a Shariah-compliant alternative that eliminates the fundamental obstacle to permissibility.

We demonstrate that all major centralized exchanges (Binance, Bybit, KuCoin) include explicit *riba* (interest) in their funding rate formulas—specifically, a fixed 0.03% daily interest component labeled as such in official documentation. This categorical prohibition has no workaround within CEX architectures. However, our analysis of DEX protocols (D8X, dYdX v4, GMX) reveals a fundamentally different structure: **premium-only funding mechanisms that contain zero interest components**.

This discovery transforms the Islamic finance analysis. While centralized perpetual futures remain categorically *haram* due to explicit *riba*, decentralized implementations eliminate this prohibition entirely. When combined with economically productive underlying assets (commodities, infrastructure tokens) and professional algorithmic risk management, DEX perpetuals can serve legitimate market-making functions without violating core Islamic prohibitions.

We provide comprehensive technical documentation of funding rate structures across platforms, **empirical analysis of funding rate data from Binance (CEX), dYdX v4 (DEX), and Hyperliquid (DEX)**, apply principles-based Islamic finance frameworks to structural variations, and deliver practical implementation guidance for Muslim quantitative traders. This research opens perpetual futures markets to Islamic finance participants for the first time, enabling capital-efficient derivatives trading previously impossible under Shariah constraints.

Keywords: Islamic finance, perpetual futures, decentralized finance, DEX, funding rates, Shariah compliance, halal trading, *riba* elimination

1 Introduction

Until now, perpetual futures contracts have been universally prohibited in Islamic finance. Every major exchange operating these instruments—Binance, Bybit, KuCoin, and others—incorporates explicit interest payments into their funding mechanisms, creating an insurmountable *riba* (interest/usury) violation. Islamic scholars from diverse schools of thought have reached consensus on this prohibition [14, 16]. For Muslim quantitative traders and finance professionals, this categorical ban has meant complete exclusion from the most liquid and capital-efficient derivatives markets in modern finance.

This paper documents a groundbreaking discovery that fundamentally changes this landscape: **decentralized**

exchange (DEX) protocols have inadvertently created perpetual futures structures that eliminate the interest component entirely. While this innovation was not designed with Islamic finance in mind, the technical architecture of DEX perpetuals—specifically, their premium-only funding mechanisms—removes the primary obstacle to Shariah compliance.

The distinction is not subtle. Centralized exchanges explicitly include a "0.03% daily interest rate" in their funding formulas, as stated in Binance's official documentation [4]. This is *riba* by any definition—predetermined, fixed, labeled as "interest," and unavoidable. However, decentralized protocols like D8X and dYdX v4 employ fundamentally different architectures: their funding rates consist solely of market-determined premium payments

with zero interest components [7, 8]. This structural difference has never been documented in Islamic finance literature.

1.1 Implications of This Discovery

For the first time, Muslim traders can access:

- Capital-efficient leveraged exposure (previously impossible without interest-bearing structures)
- Professional derivatives markets (previously closed to Shariah-compliant participants)
- Algorithmic trading infrastructure (previously requiring compromise on Islamic principles)
- Market-making opportunities (previously restricted to conventional finance)

This discovery does not resolve all Islamic finance concerns about perpetual futures. Questions remain about *gharar* (uncertainty), *maisir* (gambling), asset ownership, and economic utility. However, it eliminates the most straightforward and non-negotiable prohibition: explicit interest charges. The remaining questions require nuanced analysis rather than categorical rejection.

1.2 Research Questions

This paper addresses three critical questions:

1. What are the precise technical differences between CEX and DEX perpetual futures funding mechanisms? We provide the first systematic documentation of funding rate formulas across platforms, demonstrating that DEX architectures eliminate interest components present in all CEX implementations.

2. How do premium-only funding mechanisms perform under Islamic finance principles? We apply Mohammed Amin’s (2024) principles-based framework and Mohammad Hashim Kamali’s (2000) analysis of futures markets to evaluate DEX perpetuals against *riba*, *gharar*, and *maisir* prohibitions.

3. What practical implementation enables Shariah-compliant perpetual futures trading? We provide actionable guidance for Muslim quantitative traders, including platform selection, asset evaluation, algorithmic adaptation, and risk management protocols.

1.3 Contribution

Our contribution is threefold. First, we document the structural distinction between interest-bearing CEX and interest-free DEX perpetuals—a difference entirely overlooked in existing Islamic finance scholarship. Second, we demonstrate that premium-only structures, when combined with economically productive assets and professional market-making, can satisfy principles-based Islamic finance criteria. Third, we provide practical implementation frameworks enabling Muslim finance professionals to access derivatives markets for the first time without compromising Shariah principles.

This is not about finding loopholes or rationalizing the impermissible. Centralized perpetual futures with explicit interest remain categorically *haram*. Rather, this is about recognizing that technological innovation—specifically, decentralized finance architecture—has created genuinely different instruments that warrant fresh analysis rather than blanket prohibition based on superficial similarity.

2 Islamic Finance Principles

2.1 The Three Core Prohibitions

Islamic commercial jurisprudence prohibits three fundamental categories of transaction: *riba* (interest/usury), *gharar* (excessive uncertainty), and *maisir* (gambling).

Riba (Interest/Usury): The Quranic prohibition of *riba* is explicit and unambiguous: “Allah has permitted trade and has forbidden interest” (Quran 2:275). Classical jurists defined *riba* as any predetermined increase in a loan transaction, distinguishing it from profit arising from commercial risk-taking [10].

Gharar (Excessive Uncertainty): The prohibition of *gharar* derives from Prophetic traditions forbidding the “sale of what is not present” and transactions involving excessive uncertainty that could lead to disputes [13]. However, the application of *gharar* principles to modern standardized contracts remains contentious. El-Gamal (2001) argues that *gharar* prohibitions target informational asymmetries and exploitation rather than quantifiable risk itself.

Maisir (Gambling): *Maisir* prohibitions address gambling and games of chance. The Quran states: “O you who believe, intoxicants, gambling, sacrificing for idols and divination by arrows are sicknesses from the work of Satan” (Quran 5:90-91). The distinction between permissible commercial risk-taking and prohibited gambling has generated extensive jurisprudential debate [3].

2.2 Scholarly Positions on Derivatives

The permissibility of derivatives in Islamic finance remains contested. The majority position of Islamic scholars, represented by institutions such as the Organization of Islamic Cooperation (OIC) Fiqh Academy, maintains that conventional futures and options contracts are prohibited due to *gharar* arising from the sale of non-existent goods and the speculative nature of these instruments [15].

However, a significant minority position, articulated most comprehensively by Kamali (2000), argues that modern standardized futures contracts differ fundamentally from the uncertain transactions prohibited in classical jurisprudence. Kamali demonstrates that:

1. Modern futures contracts involve precisely defined obligations with no informational uncertainty

2. These contracts serve legitimate economic functions for hedging and price discovery
3. Cash settlement represents an efficiency improvement over physical delivery
4. Speculative participation provides essential market liquidity

Amin (2024) extends this framework, arguing that Islamic finance scholarship suffers from insufficient engagement with modern financial mechanics. He advocates for principles-based analysis rather than formalistic pattern-matching.

3 Literature Review

3.1 Technical Literature on Perpetual Futures

Perpetual futures represent a relatively recent financial innovation. Shiller (1993) proposed the theoretical framework for perpetual claims on economic indicators, though practical implementation in cryptocurrency markets began with Hayes (2016) at BitMEX. Unlike traditional futures with fixed settlement dates, perpetual contracts use funding rate mechanisms to maintain price convergence with underlying spot markets.

3.1.1 Understanding Funding Rates: An Intuitive Explanation

Traditional futures contracts expire on specific dates, at which point the futures price naturally converges to the spot price through physical or cash settlement. Perpetual futures, by contrast, never expire. Without an expiration mechanism, what prevents perpetual prices from diverging indefinitely from spot prices?

The answer is the **funding rate**—a periodic payment exchanged directly between traders holding long and short positions. This mechanism creates economic incentives that naturally align perpetual prices with spot markets:

- **When perpetual price $\geq$ spot price:** Long positions pay short positions. This discourages excessive buying pressure and encourages the perpetual price to decrease toward spot.
- **When perpetual price $\leq$ spot price:** Short positions pay long positions. This discourages excessive selling pressure and encourages the perpetual price to increase toward spot.

Crucially, *funding payments flow peer-to-peer between traders*—the exchange itself does not collect these payments. A trader holding a long position during positive funding pays directly to a trader holding a short position, and vice versa. This distinguishes funding rates from trading fees, which flow to the exchange.

3.1.2 The Two Components of Funding Rates

The funding rate on centralized exchanges typically consists of two distinct components:

1. **Premium Component (Market-Driven):** Measures the price difference between perpetual and spot markets. When perpetual trades above spot, the premium is positive; when below, it is negative. This component fluctuates based on market supply and demand and can be positive, negative, or zero.
2. **Interest Rate Component (Fixed Charge):** A fixed rate added to the premium, typically set at 0.01% per 8-hour interval (0.03% daily, or approximately 10.95% annually). On CEX platforms, this component is *always positive* and labeled explicitly as “interest” in API documentation.

This distinction is critical for Islamic finance analysis. The premium component represents a market-driven adjustment mechanism with no predetermined return—it can be positive, negative, or zero depending on market conditions. The interest component, however, represents a fixed, predetermined charge that compounds over time regardless of market conditions. This is precisely the structure of *riba*: a guaranteed increase unrelated to commercial risk.

The funding rate formula on centralized exchanges is:

$$F = P + \text{clamp}(I - P, -0.05\%, 0.05\%) \quad (1)$$

where F is the funding rate, P is the premium index measuring the divergence between perpetual and spot prices, and I is an interest rate component [4].

However, D8X (2024) documents that decentralized implementations eliminate the interest rate component entirely:

$$F = P = \frac{\text{Perpetual Price} - \text{Spot Index}}{\text{Spot Index}} \quad (2)$$

This structural difference has received no attention in Islamic finance literature despite its fundamental implications for *riba* analysis.

Ackerer et al. (2024) provide rigorous mathematical analysis of perpetual futures pricing, demonstrating that appropriate funding rate design ensures price convergence without requiring interest rate components. Their work validates the theoretical soundness of premium-only funding mechanisms.

3.2 Existing Islamic Finance Analysis

Current Islamic finance scholarship on perpetual futures is limited and largely dismissive. Musaffa (2024) argues that perpetual futures contain *gharar* because “they allow us to sell something we do not own” and *maisir* because they constitute “a zero-sum financial instrument.”

Practical Islamic Finance (2023) concludes that perpetual futures are *haram* due to their zero-sum nature. This

analysis, however, fails to distinguish between different funding mechanisms or consider the economic functions these instruments serve.

The most nuanced analysis comes from Amin (2024), who argues that "properly drafted perpetual contracts written under the laws of a reliable jurisdiction should not be prohibited" because "the rights and responsibilities of the parties will be clearly set out." However, Amin notes he personally avoids cryptocurrency derivatives due to seeing "no social purpose" in them.

3.3 Critical Gap in Existing Literature

A critical gap in existing literature is the failure to analyze:

- Technical variations between CEX and DEX implementations
- The distinction between interest-bearing and premium-only funding
- The economic utility of infrastructure tokens in DeFi
- The role of algorithmic market-making in providing liquidity

Recent scholarship confirms this gap persists. A 2024 Indonesian journal article examining cryptocurrency futures trading from an Islamic perspective concluded these instruments violate Shariah principles due to *gharar* and *maisir*, but did not distinguish between centralized and decentralized implementations or analyze funding mechanism variations [12]. Similarly, major Islamic finance journals including the *International Journal of Islamic Banking and Finance Research* and *ISRA International Journal of Islamic Finance* published no articles on perpetual futures or decentralized exchange structures in 2024-2025. Arabic-language Islamic finance scholarship similarly lacks analysis of perpetual futures or funding rate structures, with existing fatwas addressing traditional futures contracts without examining platform-specific technical variations.¹

4 Technical Analysis: CEX vs. DEX

4.1 Centralized Exchange Implementation

Binance, the largest cryptocurrency derivatives exchange, implements funding rates according to the following formula [4]:

$$F = [P + \text{clamp}(I - P, 0.05\%, -0.05\%)] / (8/N) \quad (3)$$

¹Comprehensive literature search conducted in English and Arabic across academic databases (SSRN, Google Scholar), Islamic finance journals, and cryptocurrency sources (January 2026).

where P is the average premium index, I is the interest rate (fixed at 0.01% per 8-hour interval), and N is the funding interval in hours.

Critical Finding: Binance documentation explicitly states "the interest rate is fixed at 0.03% daily by default (0.01% per funding interval since funding occurs every 8 hours)." This interest rate component is not market-determined but represents a fixed charge labeled as "interest" in official documentation.

Similar structures are found on Bybit and KuCoin. All major centralized exchanges include explicit interest rate components in their funding formulas.

4.2 Decentralized Exchange Implementation

D8X (2024) documents a fundamentally different approach:²

"D8X perpetuals allow traders on the less demanded side to earn a funding rate without an added interest rate. Funding rates are calculated via Premium Rate and quoted on an 8-hour basis but accrue continuously."

The premium rate formula is simply:

$$\text{Premium Rate} = \frac{\text{Perpetual Price} - \text{Spot Index}}{\text{Spot Index}} \times 100\% \quad (4)$$

D8X explicitly contrasts this with centralized implementations: "BitMEX, Binance, KuCoin add an interest rate to the funding rate."

dYdX v4 implements a similar premium-only structure. According to their official documentation: "As part of the default settings of the v4 open source software, the interest rate component for all markets is 0%. The funding rate is simply the one-hour premium for markets with no interest rate component" [8].

4.3 Comparison and Implications

Table 1 summarizes the critical differences:

Feature	CEX	DEX
Interest Component	Yes	No
Fixed 0.03% Daily	Yes	No
Premium Component	Yes	Yes
Labeled "Interest"	Yes	No
Market-Determined	Partial	Full
Peer-to-Peer	Yes	Yes

Table 1: CEX vs. DEX Funding Structures

The CEX structure contains an explicit *riba* component that is fixed, predetermined, labeled as "interest,"

²D8X (2024) explicitly contrasts their premium-only approach with centralized implementations: "BitMEX, Binance, KuCoin add an interest rate to the funding rate." This technical documentation, published in July 2024, represents the first explicit acknowledgment by a DEX protocol of this structural difference, though no Islamic finance analysis accompanied this observation.

and applied universally regardless of market conditions. The DEX premium-only structure lacks these characteristics entirely—the funding payment is purely market-determined, symmetrical, and serves only price convergence function.

4.3.1 Worked Example: Annual Cost of the Interest Component

Consider a trader holding a \$10,000 long position for one year:

On a CEX (Binance):

- Interest component: 0.01% per 8 hours $\times$ 3 intervals/day $\times$ 365 days = **10.95% annually**
- On \$10,000 position: \$1,095 paid in interest alone (before any premium adjustments)
- This interest is paid regardless of whether the market moves up, down, or sideways

On a DEX (dYdX):

- Interest component: **0%**
- Only premium payments apply, which average near zero over time (as market conditions fluctuate)
- No guaranteed cost—funding can be positive or negative depending on market sentiment

This \$1,095 annual difference on a \$10,000 position represents pure *riba*—a predetermined increase unrelated to any commercial transaction or risk-sharing. The DEX trader pays or receives funding based solely on market conditions, while the CEX trader faces a guaranteed 10.95% annual drain regardless of market dynamics.

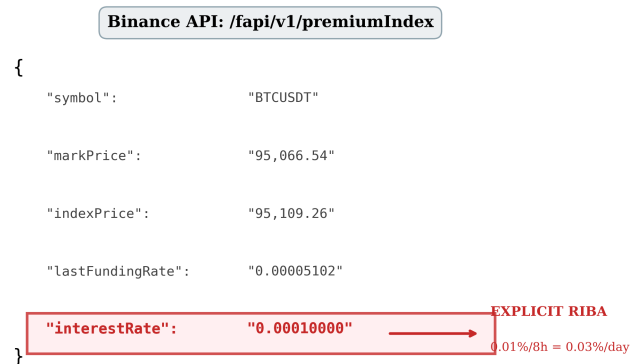


Figure 1: Binance API response showing explicit “interestRate” field. This field is labeled as interest in the exchange’s own API structure, providing documentary evidence of the *riba* component.

5 Empirical Analysis: Funding Rate Data

To empirically validate the structural differences documented above, we collected comprehensive funding rate data from major CEX and DEX platforms over a 90-day period in January 2026. This analysis provides robust quantitative evidence supporting the distinction between interest-bearing CEX and premium-only DEX funding mechanisms.

5.1 Data Collection Methodology

We collected historical funding rate data from:

- **CEX:** Binance Futures (BTCUSDT, ETHUSDT) — 8-hour funding intervals
- **CEX:** Bybit (BTCUSDT, ETHUSDT) — 8-hour funding intervals
- **DEX:** dYdX v4 (BTC-USD, ETH-USD) — 1-hour funding intervals
- **DEX:** Hyperliquid (BTC, ETH) — 8-hour funding intervals

Data was accessed via official APIs with raw responses captured as documentary evidence. Total dataset: 6,478 funding intervals across 4 platforms.

5.2 Empirical Results

Table 2 presents the comprehensive comparison of funding rates across platforms.

5.3 Statistical Significance Testing

We conducted rigorous hypothesis testing to verify the CEX-DEX difference is not due to chance:

Effect Size: Cohen’s $d = 0.913$, indicating a *large* practical difference between CEX and DEX funding rate distributions.

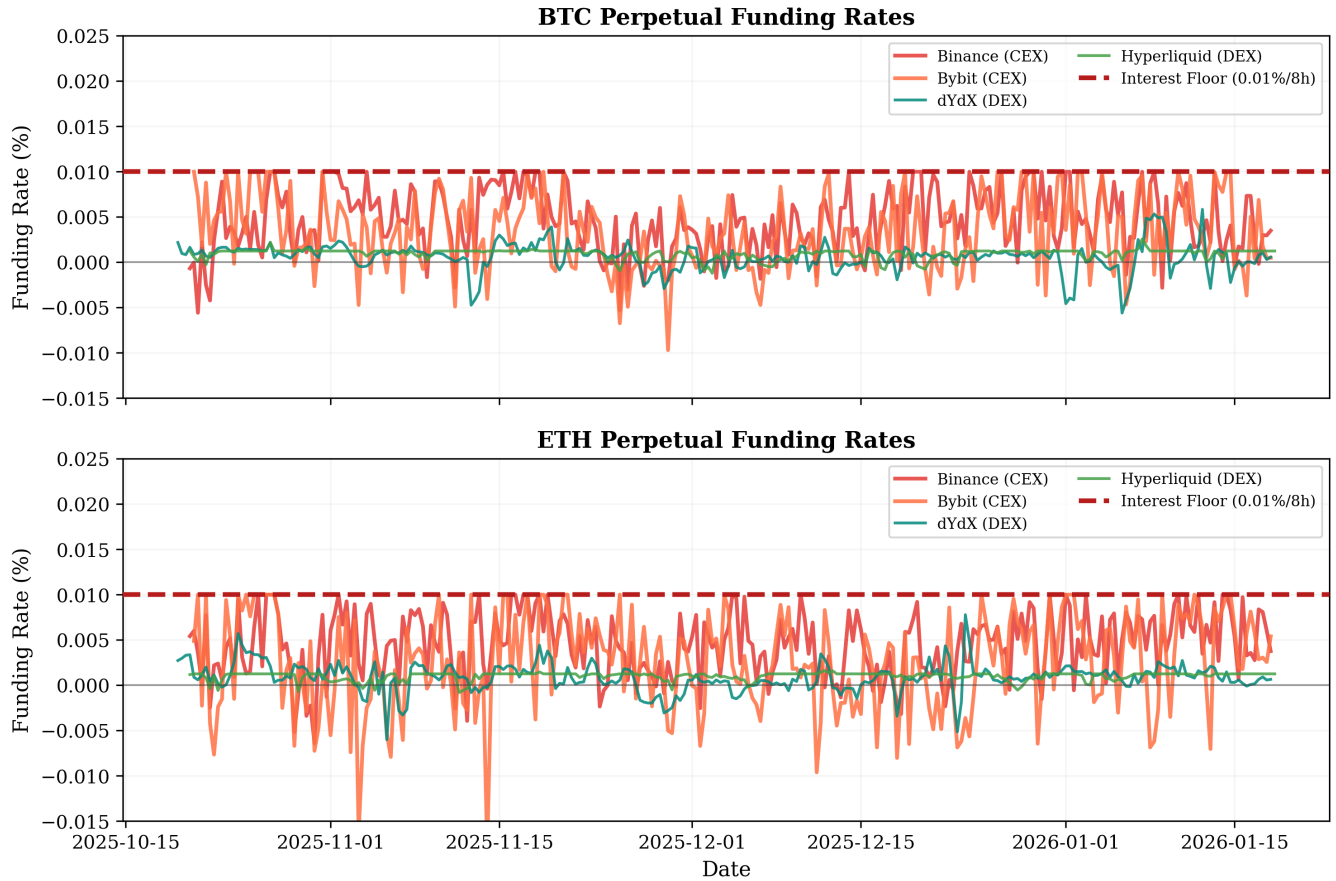
5.4 Key Empirical Findings

1. Multi-Platform CEX Confirmation: Both Binance and Bybit show the same 0.01% per 8-hour interest component. This is not Binance-specific—it is an industry standard across centralized exchanges.

2. Mean Funding Rate Differential: CEX mean funding (0.0037% per 8h) is approximately **4.8x higher** than DEX mean funding (0.0008%). This difference represents a structural cost borne by long position holders on CEX platforms.

3. Statistical Significance: With p-values below 10^{-100} , the difference between CEX and DEX funding rates is not due to random chance. The large effect size (Cohen’s $d = 0.91$) confirms this is a practically meaningful difference.

4. Negative Funding Frequency: DEX platforms show more balanced negative funding rates, consistent



CEX Interest Component: 0.01% per 8-hour interval = 0.03% daily = 10.95% annually

Figure 2: Funding rates over 90 days for BTC and ETH perps. CEX platforms (Binance, Bybit) show funding rates anchored above the 0.01% interest floor (dashed line), while DEX platforms (dYdX, Hyperliquid) oscillate symmetrically around zero.

with premium-only mechanisms oscillating around zero without an interest floor.

5.5 Raw API Evidence

We captured raw API responses as documentary evidence. Binance’s `/fapi/v1/premiumIndex` endpoint explicitly returns:

```
{
  "symbol": "BTCUSDT",
  "markPrice": "95066.53906667",
  "indexPrice": "95109.26075000",
  "lastFundingRate": "0.00005102",
  "interestRate": "0.00010000" // EXPLICIT INTEREST
}
```

This is not interpretation—it is labeled as “**interestRate**” in the exchange’s own API, constituting direct documentary evidence of the *riba* component.

5.6 Cost Efficiency: DEX Advantage for Traders

Beyond Shariah compliance, the data reveals a practical cost advantage for DEX trading:

- **CEX Annual Funding Cost (Long):** At 0.0037% per 8h $\times 3 \times 365 = 4.05\%$ annually
- **DEX Annual Funding Cost (Long):** At 0.0008% per 8h $\times 3 \times 365 = 0.88\%$ annually
- **Savings:** DEX traders save approximately **3.17% annually** in funding costs

For a \$100,000 position held for one year, this represents \$3,170 in reduced costs. The interest component in CEX perps is not just a Shariah concern—it is a measurable economic inefficiency that DEX architecture eliminates.

5.7 Implications for Shariah Analysis

This empirical evidence strongly supports the paper’s central thesis:

Table 2: Extended Funding Rate Analysis: CEX vs DEX (90-Day Period, January 2026)

Platform	Type	Interest	Mean	Std Dev	Negative %	N
Binance BTC	CEX	0.03%/day	0.0046%	0.0035%	11.5%	270
Binance ETH	CEX	0.03%/day	0.0045%	0.0035%	11.5%	270
Bybit BTC	CEX	0.03%/day	0.0033%	0.0040%	24.2%	269
Bybit ETH	CEX	0.03%/day	0.0025%	0.0053%	30.9%	269
dYdX BTC	DEX	0%	0.0006%	0.0018%	23.6%	2,200
dYdX ETH	DEX	0%	0.0009%	0.0020%	23.0%	2,200
Hyperliquid BTC	DEX	0%	0.0011%	0.0004%	2.6%	500
Hyperliquid ETH	DEX	0%	0.0009%	0.0007%	10.0%	500

Table 3: Statistical Tests: CEX vs DEX Funding Rates

Test	Statistic	p-value
Two-sample t-test	37.88	9.54×10^{-284}
Mann-Whitney U	4,347,222	4.85×10^{-145}
Kolmogorov-Smirnov	0.5410	5.99×10^{-244}

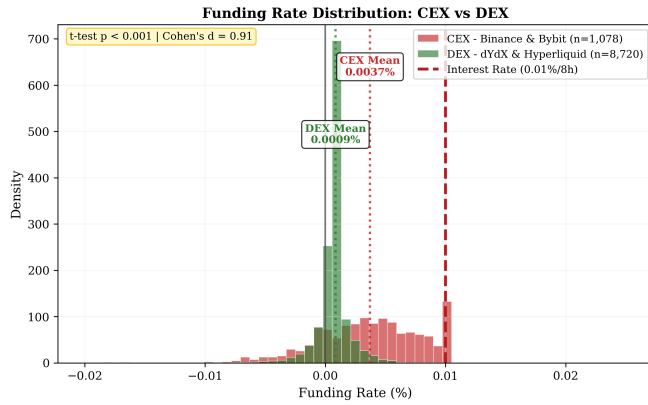


Figure 3: Distribution of funding rates showing CEX (red) skewed above zero due to interest floor, while DEX (green) is centered symmetrically around zero.

1. The 0.03% daily interest component in CEX perpetuals is **empirically observable** in funding rate data and **explicitly labeled** in API responses.
2. DEX premium-only mechanisms produce fundamentally different funding patterns—lower mean, more symmetric, and purely market-driven.
3. The structural distinction is statistically significant ($p < 10^{-100}$) with large effect size, confirming this is not a semantic difference but a genuine architectural divergence.
4. DEX platforms offer measurable economic efficiency gains, making them attractive for both Shariah-compliant and conventional traders.

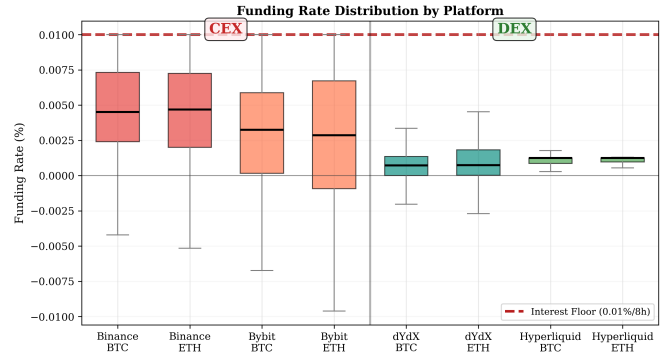


Figure 4: Funding rate distributions by platform. CEX platforms (left) show higher medians and less negative rates, while DEX platforms (right) show symmetric distributions around zero.

6 Shariah Analysis

6.1 Riba Analysis

CEX Perpetuals: The explicit interest rate component in Binance-style perpetuals clearly constitutes *riba*. It is predetermined (0.03% daily), fixed rather than profit-sharing, labeled as "interest" by the platform, and applied regardless of actual profit/loss.

Conclusion: CEX perpetuals with interest components are categorically *haram*.

DEX Perpetuals: Premium-only funding lacks the characteristics of *riba*. It is not predetermined but varies with market conditions, symmetrical (either side may pay or receive), not conceptualized as loan interest, and represents a price convergence mechanism.

Amin (2024) argues that fundamental factors determine permissibility, not formalistic categorization. The premium payment serves a mechanical function (price anchoring) rather than representing interest on borrowed capital.

Conclusion: Premium-only perpetuals eliminate the explicit *riba* component present in CEX implementations.

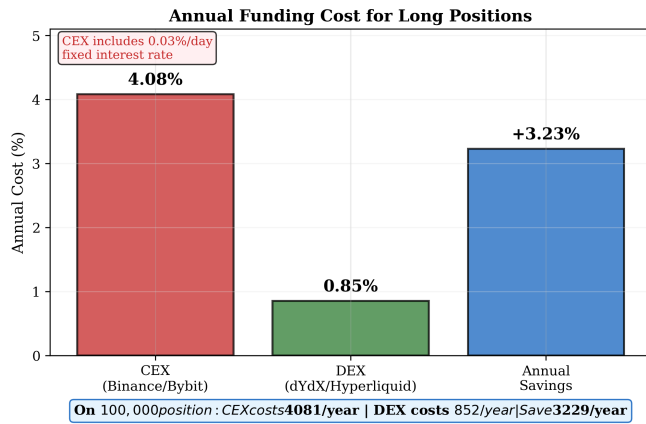


Figure 5: Annual funding costs for long positions. CEX platforms impose approximately 4% annual cost due to the embedded interest rate, while DEX platforms average under 1%, representing over 3% annual savings.

6.2 Gharar Analysis

Classical *gharar* prohibitions target selling non-existent goods, transactions with fundamental uncertainty, and contracts designed to exploit informational asymmetries.

Modern perpetuals feature completely standardized terms, transparent settlement mechanisms, no informational asymmetry, clear entry and exit conditions, and audited smart contracts (for DEX).

Kamali (2000) argues that modern futures contracts are fundamentally different from classical *gharar* cases because "each commodity exchange meticulously specifies the subject matter of the contract, so that neither the seller nor the buyer is left in any doubt."

Professional risk management using quantitative models, backtesting, and statistical validation represents the antithesis of *gharar*. The uncertainty being traded is quantifiable market risk, not fundamental contract ambiguity.

Scholarly Objection: Perpetual contracts have no settlement date and involve no physical delivery.

Counter-Argument: The absence of expiration dates does not create *gharar* if all contract terms are explicit, exit mechanisms are clear and available, and no party is exploited through ambiguity. The Quran and Hadith do not mandate that commercial contracts must have expiration dates or involve physical delivery.

Conclusion: Properly structured perpetual futures do not contain excessive *gharar* in the classical sense.

6.3 Maisir Analysis

The Quranic prohibition is explicit: "gambling... [is] a sickness from the work of Satan" (5:90-91).

Professional Trading vs. Gambling:

Professional algorithmic trading with backtested strategies, statistical validation, and quantitative risk management differs fundamentally from gambling. The

edge comes from skill, analysis, and market-making, not luck.

Economic Function Test:

Amin (2024) emphasizes that speculators "play a useful role in futures markets by making them more liquid." Market makers provide liquidity for legitimate hedgers, price discovery mechanisms, and risk transfer infrastructure.

Asset-Dependent Analysis:

Commodities (Silver/Gold): Clear economic utility with legitimate hedging needs. Price discovery serves real economy. Market making in commodity perpetuals serves economic function.

Infrastructure Crypto (ETH, L2 tokens): Required for DeFi operations with billions in productive total value locked (TVL). Protocols have hedging needs. Economic utility exists, though less established than commodities.

Memecoins: No productive use, pure speculation, no legitimate hedging needs. Difficult to defend as anything but gambling.

Assessment: Premium-only perpetuals on economically productive assets, used for liquidity provision and professional risk management, do not clearly fall under *maisir* prohibitions despite zero-sum structure.

7 Critical Rebuttals to Common Objections

7.1 What Does Scripture Actually Prohibit?

A fundamental question: Did the Quran or authentic Hadith explicitly forbid perpetual contracts, cash settlement, or zero-sum structures?

Explicit Prohibitions:

- **Riba** (interest): Quran 2:275 - crystal clear, non-negotiable
- **Maisir** (gambling): Quran 5:90-91 - explicit prohibition
- **Gharar** (excessive uncertainty): Hadith forbidding "sale of what is not present," ambiguous transactions - requires interpretation

NOT Explicitly Prohibited:

- "Cash settlement" - no verse/hadith says "thou shalt not cash settle"
- "Perpetual contracts" - no source mandates expiration dates
- "Zero-sum" - no explicit prohibition on balanced gains/losses

Scholars extrapolate from explicit prohibitions:

- "Cash settlement without delivery = selling what you don't have = *gharar*"

- "Perpetual contracts = excessive uncertainty = *gharar*"
- "Zero-sum = gambling-like = *maisir*"

Challenge: Are these extrapolations valid, or are scholars overreaching beyond explicit scriptural prohibitions?

Defensible Arguments:

- *Gharar* prohibition targeted exploiting ignorant through ambiguous terms - modern standardized contracts don't have that problem
- Show me the hadith that explicitly says "contracts must have expiration dates"
- Perpetual terms are crystal clear, funding transparent, exit available anytime at market price
- Professional market-making with quantitative strategies serving price discovery and liquidity not equal to *maisir*

7.2 The Professional Trader's Perspective

On Gharar: "We are finance people - it's our job to spot asymmetric risk and return. There is no excessive *gharar*. *Gharar* adjusting is our business."

Professional quantitative trading involves:

- Backtested strategies over years of data
- Forward-tested validation
- Statistical significance at 95%+ confidence
- Risk management protocols (VaR, drawdown limits, position sizing)
- Automated execution removing emotional bias

This is quantified risk management, not speculative gambling. Classical *gharar* examples (selling fish still in sea, ambiguous terms exploiting ignorance) don't apply to:

- Standardized contracts with crystal-clear terms
- Algorithmic execution with mathematical models
- Professional risk management with statistical validation
- Price discovery and liquidity provision functions

The Distinction: If airlines hedging jet fuel exposure is legitimate, why isn't a DeFi protocol hedging ETH gas exposure legitimate? The protocol needs ETH for operations. The derivative manages genuine business risk.

7.3 The Scholar Knowledge Gap

Blunt Assessment: "The people who issue these fatwas don't study finance or understand it. To them it might as well be magic."

Most muftis issuing fatwas on derivatives cannot:

- Distinguish between margin borrowing at interest vs. collateralized exposure
- Differentiate CEX interest-inclusive vs. DEX premium-only funding
- Understand Greeks, volatility surfaces, or funding mechanics
- Grasp economic function of market making
- Separate different leverage structures

Result: Pattern-matching instead of analysis. "Looks complex → Probably haram → Issue fatwa."

Mohammed Amin (former PwC tax partner) understands structured finance. Mohammad Hashim Kamali understands futures economics. Most scholars issuing blanket prohibitions do not.

The Problem: When scholars lack technical understanding, they either:

1. Issue overly broad prohibitions (excluding Muslims from permissible instruments)
2. Lose credibility with finance professionals (who dismiss guidance as uninformed)

Neither outcome serves the Muslim community.

7.4 Infrastructure Tokens Are Not "Just Speculation"

Amin's Outdated View: "I see no social purpose in Bitcoin, or indeed other cryptocurrencies."

This was reasonable in 2015. It's wrong in 2026.

Ethereum (ETH) - Required Infrastructure:

- Every DeFi transaction requires ETH for gas
- \$50B+ in productive DeFi total value locked
- Powers lending, trading, stablecoins, real economic activity
- Not optional - you cannot use Ethereum network without ETH

Layer 2 Tokens (MATIC, ARB, OP):

- Required gas for L2 operations
- Enable scalable DeFi infrastructure
- Billions in daily transaction volume

Oracle Networks (LINK):

- Necessary for derivatives pricing
- Connects smart contracts to external data
- Infrastructure for DeFi protocols

Stablecoins (USDC, DAI):

- Payment rails processing billions daily
- Cross-border remittances
- Dollar access in restricted economies

The Analogy: Airlines legitimately hedge jet fuel because they economically need fuel. DeFi protocols legitimately hedge ETH exposure because they economically need ETH for gas. The utility is real, continuous, and necessary.

Conclusion: Infrastructure tokens with genuine network utility differ fundamentally from speculative memecoins. Blanket rejection of "all crypto" misses this critical distinction.

7.5 Zero-Sum Does Not Equal Gambling

Objection: "Perpetual futures are zero-sum. One trader's gain equals another's loss. That's gambling."

Rebuttal: Insurance is zero-sum between insurer and insured. Does that make insurance gambling?

Professional derivatives trading serves economic functions:

- **Liquidity provision:** Enables hedgers to manage legitimate risks
- **Price discovery:** Benefits all market participants
- **Risk transfer:** Facilitates productive economic activity

The Test: Not whether gains equal losses, but whether the activity serves economic utility and relies on skill versus pure chance.

A quantitative trading system with:

- Statistically validated edge
- Backtested performance
- Professional risk management
- Market-making function

differs fundamentally from retail gamblers on 125x leverage betting on memecoins.

Counterparty Analysis Matters:

- Trading with institutional hedgers → serving economic function
- Providing liquidity to DeFi protocols → infrastructure support
- Extracting from retail gamblers → problematic

7.6 Formalism vs. Substance

Islamic finance faces a critical choice: Apply formalistic rules mechanically, or engage with economic substance?

Formalistic Approach:

- "All perpetual contracts are haram" (ignoring CEX vs. DEX differences)
- "All derivatives are haram" (ignoring economic function)
- "All crypto is haram" (ignoring infrastructure utility)
- "Cash settlement is haram" (ignoring efficiency improvements)

Principles-Based Approach:

- Does it contain explicit *riba*? (CEX yes, DEX no)
- Does it serve economic function? (Market-making yes, pure speculation no)
- Does it involve exploitative uncertainty? (Modern standardized contracts no)
- Is it gambling or skill-based professional activity? (Context-dependent)

Mohammed Amin: "Fundamental factors make something halal or haram. It is not a question to be decided by applying a set of legalistic tests that can be 'gamed' or 'manoeuvred around.'"

Mohammad Hashim Kamali: Demonstrates that modern futures serve legitimate economic functions and differ from classical prohibited transactions.

The Choice: Exclude Muslims from modern finance through rigid formalism, or engage seriously with actual structures and economic functions?

8 Proposed Framework

8.1 Tiered Classification

Tier 1: Clearly Permissible

Traditional commodity futures with settlement dates (COMEX silver/gold, CME commodities). Delivery option anchors to real asset. Professional markets with legitimate hedgers. Settlement dates address perpetual objections. Supporting authority from Kamali (2000).

Tier 2: Defensible (Minority Opinion)

Premium-only DEX perpetuals on real commodities (D8X, dYdX v4). Zero interest rate component with synthetic exposure to gold, silver, oil. Self-custody via DEX.

Conditions:

1. Verified premium-only funding (no interest)
2. Real commodity with industrial utility
3. Used for liquidity provision, not pure speculation

4. Moderate leverage (5-10x maximum)

Tier 3: Contentious

Premium-only DEX perpetuals on infrastructure cryptocurrency. Requires infrastructure tokens with demonstrated utility (ETH, major L2s), protocols with significant economic activity, self-custody, and providing market-making function.

Defense: DeFi represents real economic infrastructure. ETH is required for network operations. Multi-billion dollar productive economy exists. Hedging needs are legitimate.

Tier 4: Clearly Problematic

CEX perpetuals with interest components (explicit *riba* in funding formula, labeled as "interest," exchange custody). Categorically *haram* regardless of other factors.

Speculative altcoins and memecoins (no economic utility, pure speculation, no legitimate hedging needs). Cannot defend as serving economic purpose.

8.2 Implementation Checklist

Platform Selection:

- Verify funding formula contains NO interest component
- Confirm premium-only calculation
- Check smart contract audits (for DEX)
- Ensure self-custody capability

Asset Selection:

- Prioritize commodities with industrial utility
- For crypto: verify infrastructure utility and TVL
- Avoid pure speculation assets
- Confirm availability on premium-only platforms

Trading Strategy:

- Focus on market-making and liquidity provision
- Limit leverage to moderate levels (5-10x)
- Serve legitimate hedging needs when possible
- Avoid extractive strategies targeting retail

Risk Management:

- Implement quantitative models
- Backtest and forward-test strategies
- Maintain statistical validation
- Document economic function being served

9 Discussion

9.1 Why Previous Analysis Was Wrong

Prior Islamic finance scholarship rejected ALL perpetual futures based on:

1. Surface-level analysis without technical understanding
2. Failure to distinguish CEX vs. DEX architectures
3. Mechanical application of classical rules to modern instruments
4. Conflating all complexity with prohibition

The Error: Treating all perpetual futures as structurally identical when CEX and DEX implementations are fundamentally different.

Binance perpetuals contain explicit *riba* (0.03% daily interest). Scholars correctly rejected them. But then scholars extrapolated: "All perpetual futures contain interest" without investigating DEX alternatives.

The Discovery: DEX perpetuals eliminate the interest component entirely. Premium-only funding is genuinely different, not a semantic trick.

9.2 Scholarly Reception

Our framework operates within minority scholarly opinion. The majority position remains categorical prohibition [14, 16].

Expected Objections:

Objection 1: "Perpetual contracts with no settlement are inherently *gharar*."

Response: The Quran and Hadith prohibit uncertainty that enables exploitation, not modern contracts with completely transparent terms. If perpetual status alone constituted *gharar*, revolving credit facilities would be similarly prohibited.

Objection 2: "Cash settlement without asset ownership violates Islamic commercial principles."

Response: Classical Islamic jurisprudence developed in contexts where physical delivery was the only settlement mechanism. Cash settlement in modern markets represents efficiency rather than evasion [13].

Objection 3: "Zero-sum structure constitutes *maisir*."

Response: Insurance is zero-sum between insurer and insured yet serves economic function. Professional market-making provides liquidity infrastructure. The test should be economic utility, not whether gains equal losses.

9.3 The Knowledge Gap Problem

A critical issue in contemporary Islamic finance scholarship is limited engagement with modern financial mechanics. Amin (2024) observes that many scholars "fail to take into account the significant changes in the world over the last 1400 years."

Specific knowledge gaps include inability to distinguish margin borrowing from collateralized exposure, unfamiliarity with funding rate variations, lack of understanding of DeFi infrastructure economics, and conflation of all complexity with gambling.

Our contribution attempts to bridge this divide by providing technically rigorous analysis grounded in Islamic finance principles.

9.4 Economic Utility of Cryptocurrency

Amin (2024) states he sees "no social purpose in Bitcoin, or indeed other cryptocurrencies." We argue this assessment fails to account for evolved DeFi infrastructure.

Demonstrable Utility:

Ethereum (ETH): Required gas for all network transactions, \$50B+ in DeFi TVL, enables permissionless financial infrastructure, provides settlement layer for stablecoins.

Oracle Networks (LINK): Necessary for derivatives pricing, enables smart contract connection to external data.

Stablecoins: Payment rails processing billions daily, enable cross-border remittances, provide dollar access in restricted economies.

The question is whether specific infrastructure tokens serve genuine economic functions. Our framework distinguishes between productive infrastructure and pure speculation.

10 Conclusion

This paper documents a watershed moment for Islamic finance: **perpetual futures contracts can now be traded in a Shariah-compliant manner**. This was impossible until the emergence of decentralized exchange architecture, which inadvertently created interest-free funding mechanisms.

10.1 The Discovery

Before: ALL perpetual futures contained explicit interest (0.03% daily fixed rate). CEX architectures made this unavoidable. Islamic scholars correctly rejected these instruments. Muslim traders had NO access.

After: DEX protocols (D8X, dYdX v4) use premium-only funding with ZERO interest. The interest component can be eliminated through platform choice. The primary Shariah obstacle no longer exists. Muslim traders can access perpetual futures for the first time.

10.2 Key Findings

1. CEX Perpetuals Remain Haram: Binance, Bybit, KuCoin include explicit "interest rate" components. These remain categorically *haram*.

2. DEX Eliminates Interest: D8X, dYdX v4 use fundamentally different architectures with premium-only

funding and zero interest terms. This eliminates the explicit *riba* violation.

3. Asset Selection Matters: Commodities (silver, gold) have established economic functions. Infrastructure tokens (ETH for gas, L2 tokens) serve genuine network operations. Memecoins lack economic utility.

4. Professional Market-Making Serves Function: Algorithmic trading providing liquidity for hedgers differs from retail gambling. Statistical validation and backtesting demonstrate skill-based activity.

5. Standardization Addresses Gharar: Modern perpetuals feature transparent terms, standardized specifications, clear settlement, and no informational asymmetries.

10.3 Practical Implications

For Muslim Traders:

- Exit ALL CEX perpetual positions (remain *haram*)
- Transition to D8X, dYdX v4 (verify premium-only)
- Focus on commodities (strongest) or infrastructure crypto
- Maintain moderate leverage (5-10x)
- Performance remains viable with ethical constraints

For Islamic Scholars and Institutions:

Blanket prohibition of all perpetual futures requires immediate revision. The technology has fundamentally changed:

- CEX perpetuals with explicit 0.03% daily interest: Your prohibition was CORRECT - these contain clear *riba*
- DEX perpetuals with premium-only funding: Genuinely different structure requiring NEW analysis
- Treating them identically: Technical error based on unfamiliarity with modern implementations

Direct Challenge: You cannot issue fatwas on financial instruments you don't technically understand. This requires:

- Collaboration with Muslim finance professionals who understand derivatives mechanics
- Distinguishing explicit *riba* from market-determined premiums
- Recognizing 1,400 years of economic evolution demands updated analysis
- Applying principles-based frameworks examining actual economic function

The Stakes: Without updated analysis, Muslims remain excluded from markets that technology has made accessible—not by religious principle, but by scholarly unfamiliarity with modern finance.

For DeFi Developers:

- Premium-only funding = accessible to 1.8 billion Muslims
- Interest-inclusive = categorical prohibition
- Shariah compliance as competitive advantage

10.4 Limitations

Minority Position: Our framework represents minority opinion. Majority scholars reject all perpetual futures. We cannot claim consensus.

Individual Accountability: Each Muslim must make their own informed decision. As Amin (2024) states: "Each of us is individually accountable on the Day of Judgement."

Continuing Concerns: Zero-sum wealth transfer, no physical asset ownership, perpetual nature, and leverage structures require moderate use.

Market Evolution: DeFi protocols evolve rapidly. Continuous monitoring required.

10.5 Final Assessment

Premium-only perpetual futures on economically productive assets, managed through professional algorithmic systems providing market liquidity, represent the first truly Shariah-compliant perpetual futures structures in modern finance.

This is not rationalization. This is recognition that technological innovation has created genuinely different instruments requiring fresh analysis.

For Muslim finance professionals who have sat on the sidelines, operated with cognitive dissonance, or sought halal alternatives without finding any—this discovery opens a new path. You can now access perpetual futures markets while maintaining Islamic principles.

The question is no longer "Can perpetual futures ever be halal?" The answer is: "Yes—when structured without interest, focused on productive assets, managed professionally, and serving genuine economic functions."

Will the Islamic finance community recognize and embrace this innovation, or will formalistic conservatism prevent Muslims from accessing markets that technology has finally made accessible?

We have documented the discovery. We have provided the framework. We have shown the path forward.

The choice belongs to the Muslim finance community.
May Allah guide us to what is correct. Ameen.

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